1. MINING

1.1. Mining factors and techniques

Mining is the extraction of rocks and minerals from the ground. It depends on certain **natural factors**, especially the **existence of large underground deposits**. The **depth, size and quality** of these deposits determine if mining is **technically possible and profitable**.

There are several **techniques** used to extract minerals:

- **Opencast mining:** Also called **open-pit mining**, this method extracts mineral deposits close to the surface. It creates large holes in the ground. Smaller holes used to extract rocks are called **quarries**.
- **Underground mining:** This technique is used for **deep deposits**. It needs **access pits**, **ventilation**, **drainage pipes** and **galleries** to reach the minerals safely.
- **Drilling:** This method uses **drilling rigs** to extract **oil and gas** from the Earth's subsurface. When carried out at sea, the rigs are installed on **offshore platforms**.

Human factors also affect mining. The **laws of a country** can restrict mining activities that damage the **natural environment**, for example, to protect forests or prevent pollution.

1.2. Mining production around the world

There are **three main groups** of minerals and rocks extracted by mining:

- **Metallic minerals:** These include **iron, copper and aluminium**, which is obtained from **bauxite**, common in tropical regions. **Gold, silver and mercury** are also metallic minerals with high economic value.
- **Non-metallic minerals and rocks:** These are used in **industry and construction**. For example, **phosphates, nitrates and potassium salts** are important for the chemical and pharmaceutical industries, while **clay, granite, slate and marble** are used in construction. Modern technology industries need **coltan, lithium and diamonds** for electronic devices and batteries.
- **Minerals used for energy:** These include **coal, petroleum (crude oil)** and **radioactive minerals** such as **uranium and radium**, used to produce **nuclear energy**.

2. ENERGY SOURCES

Energy sources provide the power needed for **industry, transport, homes and offices**.

They can be classified into **primary** and **secondary sources**:

- **Primary sources** remain in their natural state, such as **sunlight, wind, water and minerals**.
- **Secondary sources** are transformed from primary ones, for example, **electricity and fuels**.

A **fuel** is any material that releases energy when it reacts, such as **fossil fuels** (coal, oil and gas). **Electrical energy** is an **intermediate form** because it is produced from primary energy sources but

can later be converted into **thermal, luminous or mechanical energy**.

2.1. Non-renewable energies

These come from **sources that cannot be replaced quickly** after being used. They include **fossil fuels**, **metal ores**, and **radioactive minerals**. These sources have powered most of the world's industry for centuries.

- **Radioactive minerals: Nuclear energy** is produced by **fission**, which splits the atomic nucleus of uranium, plutonium or thorium. This process takes place in **nuclear power stations**, producing large amounts of electricity but creating **radioactive waste**.
- **Coal:** A solid mineral fuel obtained through mining. It is used in **thermal power stations** to produce electricity and as a basic raw material for **iron and steel industries**. Although coal was one of the first energy sources of the Industrial Revolution, it is now less used because it is **expensive and highly polluting**.
- **Natural gas:** Found near **oil fields**, it is mainly used in **thermal power plants** to generate electricity. It produces less pollution than coal or petroleum, so its **consumption is increasing**.
- **Petroleum (crude oil):** The **most important energy source** in the world. It is used to produce **fuels**, **electricity** and **petrochemical products** such as plastics and fertilisers. Petroleum is extracted from **underground or undersea reservoirs** and transported through **pipelines or oil tankers**.

2.2. Renewable energies

Renewable energy sources are **natural resources that do not run out**, or that regenerate quickly. They are often called **alternative energies** because they replace conventional non-renewable ones.

- **Wind energy: Wind turbines** transform wind into electricity. They can be small turbines for homes or large ones that form **wind farms** on land or at sea (**offshore**).
- **Water energy: Hydroelectric power** uses the force of moving water stored in a **reservoir**. Water flows through pipes to turn **turbines**, generating electricity. **Tidal and wave energy** also use the movement of the sea.
- **Solar energy: Solar panels** capture the Sun's energy. **Photovoltaic panels** produce electricity, while **solar thermal panels** produce heat. Solar energy can be generated in **small domestic systems** or in **large solar farms**.
- **Geothermal energy:** Produced by the **heat inside the Earth**. Underground water and steam reach very high temperatures and are extracted to produce electricity.
- **Biomass and bioenergy:** Obtained from **organic waste and crops** such as wood, straw, manure, corn or soya. It can produce **heat, electricity and biofuels** for vehicles.

Renewable energies are cleaner and help reduce **pollution and climate change**, although their installation requires investment and technology.

3. INDUSTRIAL ACTIVITY AND ITS EVOLUTION

3.1. Evolution of industry

The way in which humans have transformed raw materials into products has changed over time, passing through different **industrial stages**:

- **Craftsmanship:** Before the Industrial Revolution, all goods (clothes, shoes, food, ceramics, metal tools, etc.) were made **by hand** by craftsmen in small workshops. Each craftsman carried out the entire process, and every product was **unique**.
- **Industrial Revolution (18th century):** Mechanisation and the use of **machines powered by coal** replaced manual production. Goods began to be made in **factories**, and production became **faster and cheaper**. Workers specialised in certain tasks.
- **Second Industrial Revolution (late 19th century):** The appearance of **production lines** and the use of **electricity** made production more complex. Factories grew, and each worker had a **specific role**, often repeating the same task.
- **Third Industrial Revolution (20th century):** Technological and computer advances increased productivity. **Automation and robotics** reduced the need for manual labour and allowed the creation of more sophisticated products.

3.2. Classifying industries

Industries can be classified in several ways:

By the type of goods and market:

- **Heavy industry:** Transforms raw materials into **semi-finished products** used by other industries. These industries require **large capital investments**, occupy **vast areas**, and are often located near **energy sources and raw materials**.
  Examples: **chemical, petrochemical, cement, steel and energy-producing industries**.
- **Capital goods industry:** Uses products from heavy industry to make **machinery, materials and equipment** for other sectors. Requires large factories, high investment and a **skilled workforce**.
  Examples: **machine, construction, aeronautical, aerospace, shipbuilding and railway industries**.
  - **Construction sector:**
    - **Private projects:** Houses, offices, workplaces.
    - **Public projects:** Dams, ports, roads, railways, airports, etc.

- **Consumer goods industry:** Produces goods directly for consumers, using raw or semi-finished materials. Examples include **food, textiles, consumer electronics, cosmetics, cars, furniture and light chemical products** (detergents, paints, plastics).

By the technology used:

- **High-technology industries:** Use advanced science and innovation (computing, telecommunications, electronics, biotechnology).
- **Mature industries:** Have reached their maximum development; their technology is **outdated**, for example, **metal industries, shipbuilding and basic manufacturing**.

4. LOCATION OF INDUSTRY AND OFFSHORING

4.1. Factors of location

The **location of industry** means choosing a place where **production and supply costs** are as low as possible.

Main factors include:

- **Natural factors:** Proximity to **raw materials and energy sources** reduces transport costs. Example: the **fish-canning industry** near ports.
- **Infrastructure:** Good **transport, communication and energy networks** are essential to receive materials and distribute products.
- **Workforce:** Some industries need **skilled workers**, others need a **large, cheap labour force**.
- **Legislation and government:** Environmental laws and **government aid** influence industrial location. Some governments promote **industrialisation** by investing in infrastructure and training workers.
- **Proximity of other industries:** Helps share services and benefit from existing infrastructure.
- **Business objectives:** Entrepreneurs also consider **personal preferences** and **economic expectations**.

4.2. Location of industry around the world

The main industrial regions are in **developed countries** that industrialised between the **Industrial Revolution** and the **20th century**.
Since the 1980s, due to **cheaper transport, better communications and globalisation**, many industries have moved to **developing countries**. This process is called **offshoring**.
Developing countries attract industries by offering **tax reductions**, **cheap labour**, and **less strict environmental laws**. This reduces production costs for companies but can create **pollution and social problems**.
Offshoring allows companies to produce goods more cheaply and sell them worldwide, but it also causes **job losses** in developed countries and **unequal development** between regions.